

Computer Forensics –

1. Overview of Computer Forensics

Definition

Computer Forensics (also known as Digital Forensics) is the process of identifying, collecting, preserving, analyzing, and presenting digital evidence from computers, mobile devices, networks, and storage media in a legally acceptable manner.

It helps investigators determine:

- What happened?
- When did it happen?
- Who performed the action?
- How was the attack conducted?
- What data was affected?

Real-Time Example

A company discovers that confidential customer data has been leaked.

A forensic investigator:

1. Seizes the suspect's laptop.
 2. Creates a forensic image (exact copy).
 3. Examines browser history, emails, and USB usage logs.
 4. Finds evidence that the employee copied files to a USB drive.
 5. Generates a report for legal action.
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Objectives of Computer Forensics

1. Recover deleted data.
2. Investigate cyber crimes.
3. Identify attackers.
4. Preserve evidence for court.

5. Analyze security incidents.
 6. Support law enforcement investigations.
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Areas of Computer Forensics

1. Disk Forensics

Investigates hard disks, SSDs, USB drives.

Example:

Recover deleted project files from an employee's laptop.

2. Network Forensics

Analyzes network traffic.

Example:

Identify the source IP address of a DDoS attack.

3. Mobile Forensics

Investigates smartphones and tablets.

Example:

Recover deleted WhatsApp messages.

4. Memory Forensics

Analyzes RAM contents.

Example:

Find malware running only in memory.

5. Email Forensics

Examines emails and attachments.

Example:

Track phishing emails sent to employees.

2. Difference Between Computer Crime and Unauthorized Activities

Computer Crime	Unauthorized Activity
Illegal activity punishable by law	Violation of company policy
Hacking bank server	Using office PC for personal work
Data theft	Installing unauthorized software
Legal action possible	Administrative action possible

Examples of Computer Crime

Example 1: Bank Hacking

A hacker steals 50 lakh from online banking accounts.

Result: Criminal offense under cyber laws.

Example 2: Ransomware Attack

An attacker encrypts company data and demands payment.

Result: Cyber crime.

Examples of Unauthorized Activities

Example 1: Employee Accessing HR Files

An employee accesses salary records without permission.

Result: Unauthorized activity.

Example 2: Personal Software Installation

An employee installs games on office computers.

Result: Policy violation.

Comparison Table

Parameter	Computer Crime	Unauthorized Activity
Legality	Illegal	May not be illegal
Punishment	Court/legal action	Warning/suspension
Intent	Malicious	Often careless
Example	Data theft	Accessing restricted folder

3. Cyber Laws

Cyber laws are legal rules governing activities performed using computers, networks, and the internet.

In India, cyber laws are mainly governed by:

Information Technology Act 2000

The IT Act provides legal recognition to:

- Electronic records
 - Digital signatures
 - E-commerce transactions
 - Cyber crime investigations
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Important Sections of IT Act 2000

Section 43

Unauthorized access, downloading, copying data.

Example:

Employee copies company database without permission.

Section 66

Computer-related offenses and hacking.

Example:

Hacker gains access to a server and modifies data.

Section 66C

Identity Theft

Example:

Using another person's Aadhaar or login credentials.

Section 66D

Online Fraud

Example:

Fake bank call asking for OTP.

Section 67

Publishing Obscene Content

Example:

Uploading illegal content online.

Section 72

Breach of Confidentiality

Example:

System administrator leaks customer information.

Real-Time Cyber Law Case

ATM Malware Attack

A cybercriminal installs malware on ATM systems.

- Evidence collected from hard disks.
- Logs analyzed.
- Malware traced to attacker.

Relevant sections:

- IT Act Section 66
 - IPC Fraud Provisions
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4. Process of Computer Forensics (Six Steps)

The forensic process consists of six major phases.

1. Identification

Recognize potential evidence sources.

Activities

- Identify computers
- Mobile phones
- Servers
- USB devices
- Cloud accounts

Example

A company suspects insider data theft.

Investigators identify:

- Employee laptop
 - USB drive
 - Email account
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2. Preservation

Protect evidence from alteration.

Activities

- Disconnect systems
- Use write blockers
- Create forensic images

Example

Police seize a laptop and immediately clone the disk to prevent data modification.

3. Collection

Gather digital evidence.

Activities

- Disk imaging
- Log collection
- Memory dump acquisition

Example

Collect:

- Firewall logs
 - Browser history
 - USB activity logs
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4. Examination

Extract useful information.

Activities

- Recover deleted files

- Analyze metadata
- Search keywords

Example

Recover deleted Excel files containing confidential customer records.

5. Analysis

Determine what happened.

Activities

- Timeline analysis
- Event correlation
- Attack reconstruction

Example

Investigator discovers:

- USB inserted at 2 PM
- Files copied at 2:05 PM
- User logged out at 2:10 PM

Conclusion: Data theft occurred.

6. Reporting / Presentation

Prepare findings for management or court.

Report Includes

- Evidence collected
- Tools used
- Findings
- Screenshots
- Conclusions

Example

Court receives:

- Timeline
- Log analysis
- File recovery evidence

to prove insider theft.

Diagram of Forensic Process

Identification

Preservation

Collection

Examination

Analysis

Reporting

5. Need for a Forensics Investigator

A forensic investigator is required because cyber attacks often leave digital evidence that must be collected and analyzed professionally.

Responsibilities of Forensics Investigator

1. Collect Digital Evidence

Example:

Recover deleted company documents.

2. Preserve Evidence Integrity

Example:

Create SHA-256 hash before analysis.

3. Investigate Security Incidents

Example:

Analyze ransomware infection source.

4. Support Law Enforcement

Example:

Provide evidence during court proceedings.

5. Identify Attackers

Example:

Trace attacker IP addresses from logs.

6. Recover Lost Data

Example:

Recover accidentally deleted project files.

Real-Time Scenario

Insider Data Theft Investigation

Employee resigns and joins competitor.

Company suspects data theft.

Forensics investigator:

1. Creates forensic image.
2. Recovers deleted files.

- 3. Examines USB logs.
 - 4. Finds copied confidential documents.
 - 5. Generates report.
 - 6. Evidence used in court.
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Tools Used in Computer Forensics

Tool	Purpose
FTK Imager	Disk Imaging
Autopsy	Disk Analysis
EnCase	Forensic Investigation
Wireshark	Network Analysis
Volatility	Memory Forensics
Cellebrite	Mobile Forensics
Magnet AXIOM	Digital Investigation
